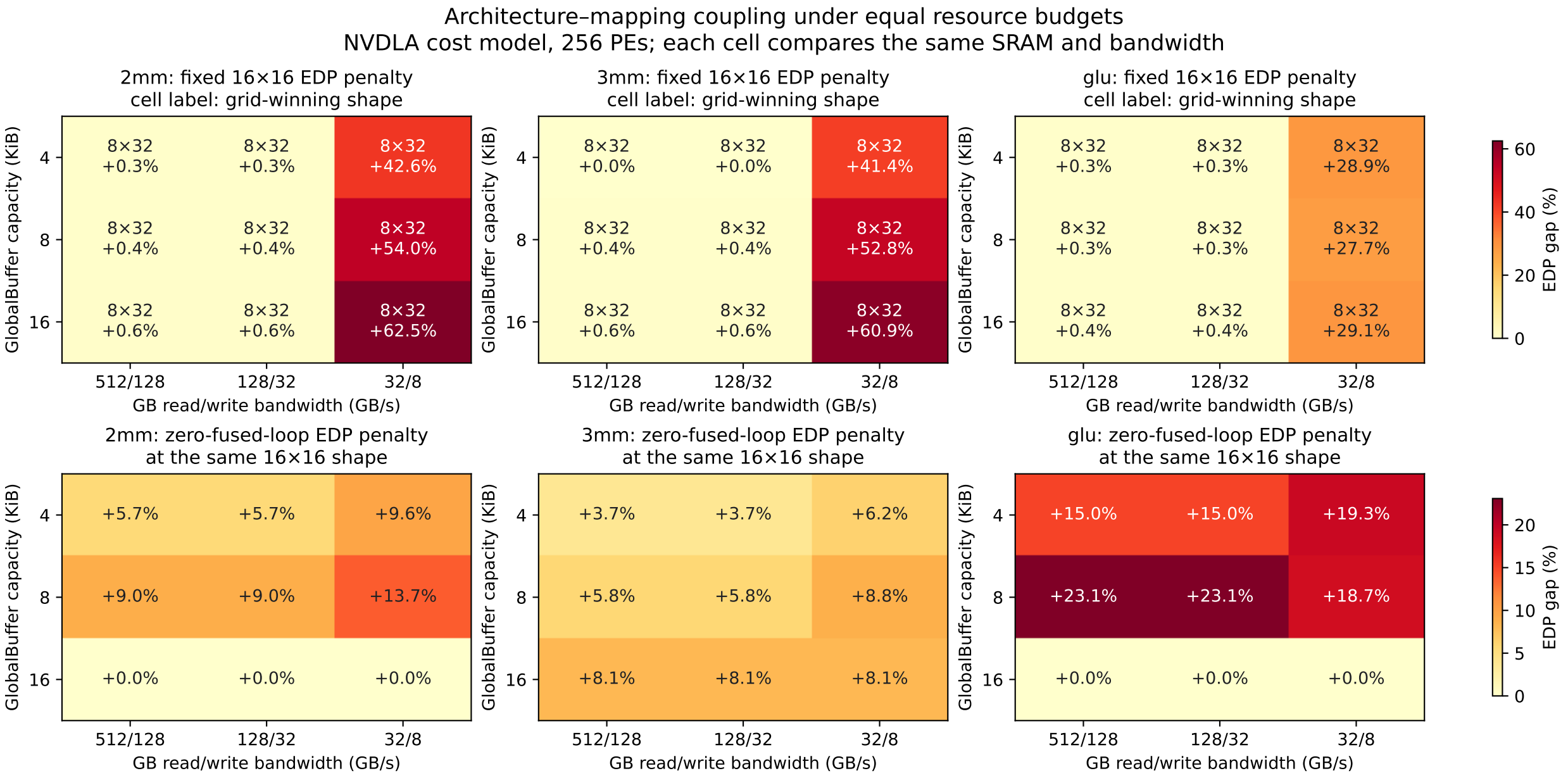
Architecture–mapping coupling under equal resource budgets
NVDLA cost model, 256 PEs; each cell compares the same SRAM and bandwidth
2mm: fixed 16×16 EDP penalty
cell label: grid-winning shape
3mm: fixed 16×16 EDP penalty
cell label: grid-winning shape
glu: fixed 16×16 EDP penalty
cell label: grid-winning shape
GlobalBuffer capacity (KiB)
4
8
16
8×32 +0.3%
8×32 +0.3%
8×32 +42.6%
8×32 +0.4%
8×32 +0.4%
8×32 +54.0%
8×32 +0.6%
8×32 +0.6%
8×32 +62.5%
8×32 +0.0%
8×32 +0.0%
8×32 +41.4%
8×32 +0.4%
8×32 +0.4%
8×32 +52.8%
8×32 +0.6%
8×32 +0.6%
8×32 +60.9%
8×32 +0.3%
8×32 +0.3%
8×32 +28.9%
8×32 +0.3%
8×32 +0.3%
8×32 +27.7%
8×32 +0.4%
8×32 +0.4%
8×32 +29.1%
512/128
128/32
32/8
GB read/write bandwidth (GB/s)
EDP gap (%)
0
20
40
60
2mm: zero-fused-loop EDP penalty
at the same 16×16 shape
3mm: zero-fused-loop EDP penalty
at the same 16×16 shape
glu: zero-fused-loop EDP penalty
at the same 16×16 shape
+5.7%
+5.7%
+9.6%
+9.0%
+9.0%
+13.7%
+0.0%
+0.0%
+0.0%
+3.7%
+3.7%
+6.2%
+5.8%
+5.8%
+8.8%
+8.1%
+8.1%
+8.1%
+15.0%
+15.0%
+19.3%
+23.1%
+23.1%
+18.7%
+0.0%
+0.0%
+0.0%
0
5
10
15
20